5G MAG The Media Connectivity Association



Bridging Standards and Deployments

At the intersection of Media and Connectivity, for generations to come

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5G-MAG Chair

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STREAMING – PRODUCTION – BROADCAST – REAL-TIME COMMS – AVATAR – EXTENDED REALITY
5G/6G – NON-TERRESTRIAL NETWORKS – DATA COLLECTION – NETWORK APIs – COMPUTING – AI
CONNECTED DEVICES – VIRTUAL WORLD – VOLUMETRIC VIDEO – CODECS – PROTOCOLS – AUDIO



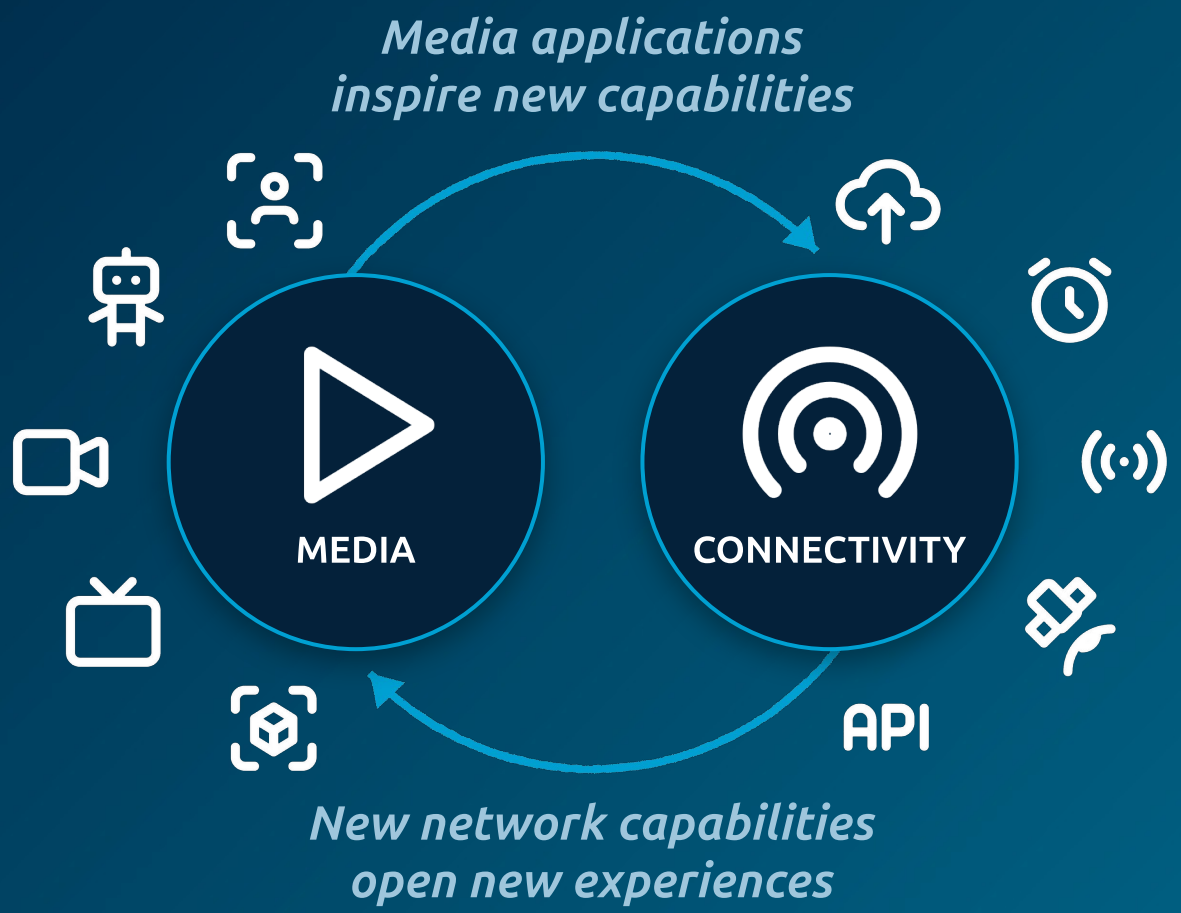




SCOPE

At the intersection of Media and Connectivity

- Fast tech evolution
- Global ecosystem
- Built better together
- Solution to pain-points



Connected Devices
Streaming Platforms
Next Generation Codecs
Internet-based Protocols
eXtended Reality
Wireless Video
Real-Time Comms
Data and Analytics
Content Production
Satellite
Network APIs
Artificial Intelligence
Quality of Service
Latency
Energy Consumption
Synchronization
Spatial Media
...

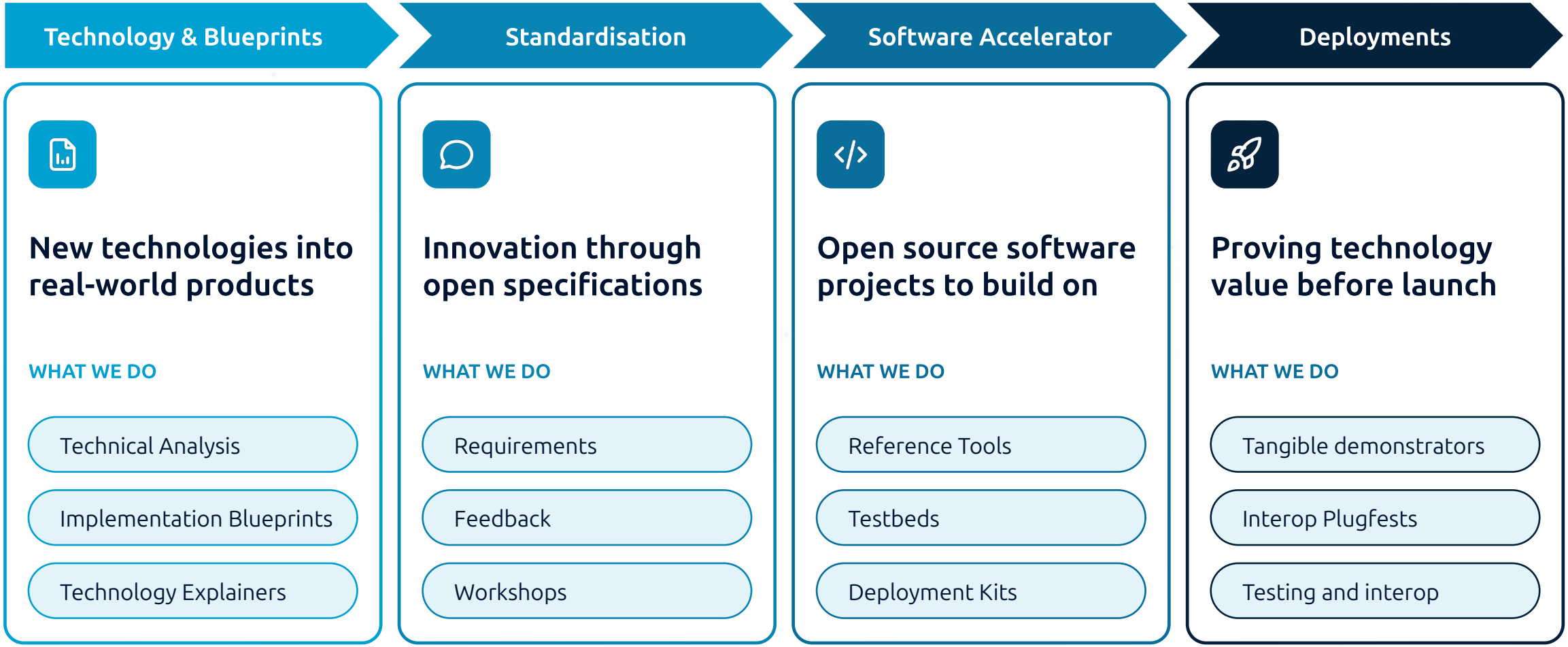
From open specifications to real-world deployments



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RUNNING THE LOOP

From open specifications to real-world deployments



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WHAT WE BUILD

19 projects across 7 technology areas

Content Delivery and Streaming

- 5G Media Streaming (5GMS)
- UE Data Collection, Reporting and Event Exposure
- Video Operation Points
- DVB-I Services over 5G Systems
- Content Delivery Protocols and Tools

Immersive Media

- XR/3D Scenes with MPEG-I Scene Description
- MPEG V3C Immersive Platform
- Dynamic Mesh Coding
- Beyond 2D Video Experiences
- Conversational Avatar Communication with MPEG ARF

Broadcast and Multicast

- 5G Broadcast - TV and Radio Services
- 5G Broadcast - Emergency Alerts
- 5G Multicast Broadcast Services (MBS)

Real-Time Communications (RTC)

- Real-time Media Communication (RTC) Architecture

Connected Media Production

- CAMARA Connectivity Quality Management APIs
- Non-Public Networks
- Time-Sensitive Communications (TSC)

Non-Terrestrial Networks (NTN)

- Non-Terrestrial Networks in 5G Systems

Towards 6G Media

- AI in Media Services

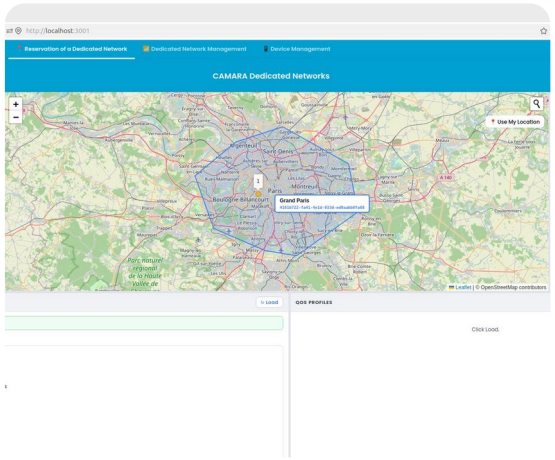
WHAT WE BUILD

The technology in action, in numbers









13
Reference Tools, open-source software

3
Testbeds and evaluation frameworks

2
Validation Tools, conformance checks

68
repositories: 47 public, 21 early access

207
issues raised on specifications

35
inputs and liaison statements to SDOs

More information at **5g-mag.com/developer** · **/reference-tools** · **/testbeds** · **/early-access**

JOIN THE EFFORT

What you get that you won’t get alone

We offer a neutral framework to collaborate within minimal bureaucracy, focused on results.



Shape standards early

Shape the standards and technologies before others set them without you.



Early access to pre-public code

Access implementations before publication and be faster to market.



Prepare for plugfests & interop

Test your implementation in advance and build reference vector traces.



Mutualised effort to grow your project

From idea to reference code to demo and trial, on shared effort.



Exposure at events, demos and trials

Build once, together, instead of funding the same work alone.



De-risk on deployments

Back technologies that actually get deployed, not specs that stall.

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Going deeper

Technology, Standardisation, Software Accelerator, Deployments, Membership

Technology & Blueprints

Understand it. Prove its value. Scale it.

Specification analysis for real-world applications

Making standards actionable

Lowering the barrier to entry

Technical Analysis

Understanding how specifications support application requirements.

Implementation Blueprints

Turn standards into implementation-ready blueprints.

Examples of **Standards Development Organizations (SDO)**

3GPP A GLOBAL INITIATIVE · MPEG · WAVE · IETF · DASH Industry Forum · DVB · CAMARA THE TELCO GLOBAL API ALLIANCE · KHRONOS GROUP

Standardisation Activities

Advancing standardisation with implementation and experience-based feedback

Active engagement with standards bodies

Requirements based on application needs

Identification of bugs and enhancements

Requirements towards SDOs

Requirements towards standards bodies, backed by industry inputs and surveys.

Industry Surveys

Share your input to help shape the requirements 5G-MAG brings to standards bodies.

Feedback to SDOs

Per-SDO feedback tracked on GitHub, and how an issue is carried through to a correction or liaison input.

Liaison Statements & Inputs

Tables of LS sent to and received from SDOs.

Workshops for Standards

The workshops that fed directly into 3GPP study and work items.

Standardisation activities in numbers

207

issues raised on specifications through GitHub

3GPP

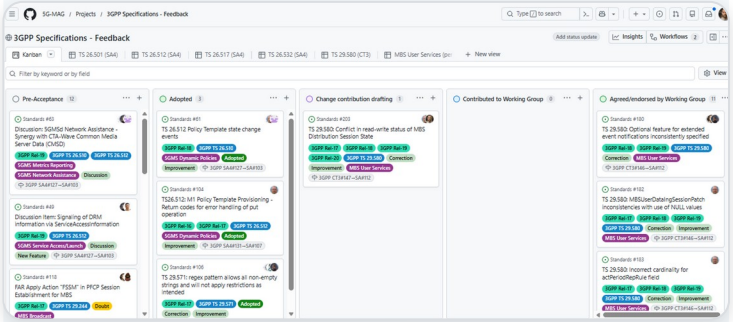
engagement as Market Representation Partner

35

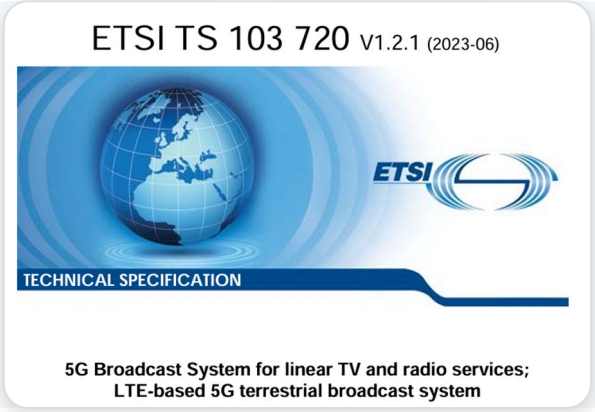
inputs and liaison statements to SDOs

6

presentations at 3GPP requirements workshops



Issues found on specifications through GitHub



Profiles of Standards Specifications



Requirements towards Standards Bodies

Media Connectivity Software Accelerator

Early feedback and validation

Software projects & Dev community

No duplication and fast deployment

Open by design and IPR-friendly

Reference Tools

Open-source implementations of the specifications.

Testbeds and Evaluation Frameworks

Environments to measure or evaluate something.

Deployment Kits

Packages, images and apps to put into a product. Coming soon.

Enabling open-source software from the start of every project

	TODAY: OPEN SOURCE AFTER THE FREEZE	WITH OPEN SOURCE FROM THE START
Evaluation	**Duplicated prototypes.** Each organization builds its own to evaluate a draft; results differ and feedback is slow.	**One shared baseline.** Drafts are validated against running code before the freeze, while gaps are cheap to fix.
Interoperability	**Late interop bugs.** Testing starts from scratch at plugfests; errors, some in the specification, surface late.	**Faster plugfests.** Implementers start from the same code: shorter preparation, faster root-cause analysis.
Deployment	**Slow deployment.** Years can pass from a frozen specification to production-ready code; early adopters carry the cost alone.	**Earlier deployment.** End-to-end demos and trials run before commercial products, and feed evidence back into the standard.

1 Plan it at launch → 2 Engage implementers during drafting → 3 Make the value tangible

"Open-source software and evaluation platform: home and scope identified at launch."

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Software Accelerator in numbers

19

projects across media and connectivity

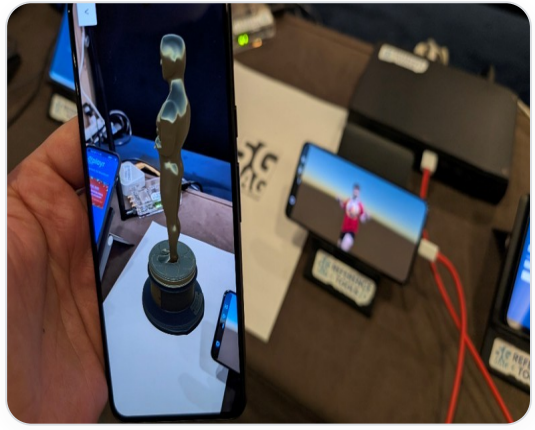


68

project repositories, public and early access

+20,000

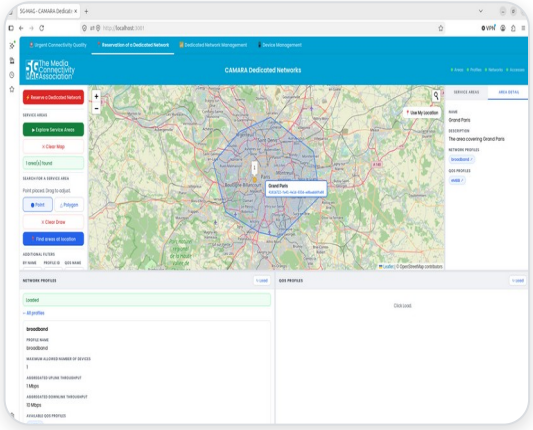
clones: unique cloners per day





Monthly

Dev Public Call and rolling releases



Growing community of developers

Organizations contributing to the Software Accelerator

BBC

Big Blue Marble

Bitstem

Dolby Laboratories

EBU

Ericsson

Fraunhofer FOKUS

Huawei

idTOLU

InterDigital

iTEAM-UPV

Motion Spell

Nokia

Philips

phine.tech

Qualcomm

Resillion

Sofia Digital

A note on the Open-Source Software Licenses

Protecting IPR. Enabling industry collaboration.

What is the License Model?

1. The 5G-MAG Public License v1.0 is a modified version of Apache 2.0 which allows:
   - **Free Non-Commercial Use:** use, copy, modify and distribute for study, academic research, testing and validation.
   - **Commercial Exploitation:** use the software in commercial services, trials and live deployments, but:
     - **FRAND Patent Licensing:** essential contributor patents are made available under **FRAND** (Fair, Reasonable, and Non-Discriminatory) terms.
2. Repositories with software derived from other licenses keep those licenses.

How to Contribute?



1. Sign an Individual or Corporate Contributor License Agreement (CLA).
2. Anybody can contribute: no 5G-MAG membership needed.
3. Members get priority support from the Project Office, and define where resources go.

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Deployments: demos, interop plugfests and applications

Specifications to code. Code to deployments.

Make value proposition tangible to the industry

Interoperability through early testing and plugfests

Supporting application development and trials

End-to-End Demos and Use Cases

User-facing demonstrations that abstract technical complexity.

Interoperability Events and Plugfests

With shared code, so interop does not start from scratch at plugfests.

Showcase at Industry Events and Trials

Promotion and demonstration of value, not just specs.

Where to meet us

MWC Barcelona

Recurrent large event in Barcelona, with a 5G-MAG stand.

IBC

Recurrent large event in Amsterdam, with Reference Tools demos.

Future Media Townhall

Yearly conference at IBC, co-organized by 5G-MAG and IEEE BTS.

OSCAR Workshop

Open-Source Core, Applications and RAN.

OSMART Workshops

Open-Source Media Application Reference Tools.

Dev Public Call

Open monthly session, no membership needed. Last Friday of the month, 13:00 CET/CEST.

5G MAG The Media Connectivity Association



Join, collaborate, sponsor software development & scale your projects

www.5g-mag.com/membership

JOIN THE EFFORT

Addressing industry pain points with open specifications

Member-driven

we add resources to their efforts

Open

to the industry to engage, collaborate

Agile

at the pace of standards and market

Pragmatic

with actionable activities & outcomes

3

WORKGROUPS focused on **TECHNOLOGY AND STANDARDS**, **DEVELOPMENT AND ECOSYSTEM**, and **PROMOTION AND COMMUNICATION**

JOIN THE EFFORT

Our current membership



More information at **5g-mag.com/membership**

JOIN THE EFFORT

Membership information and Fee Structure

Membership Fee Category	Annual Revenue	Annual Membership Fee
LARGE	**> €1 billion**	**€ 15,000**
Membership Fee Category	Annual Revenue	Annual Membership Fee
MEDIUM	**€100 million – €1 billion**	**€ 10,000**
Membership Fee Category	Annual Revenue	Annual Membership Fee
SMALL	**€5 million – €100 million**	**€ 5,000**
Membership Fee Category	Annual Revenue	Annual Membership Fee
MICRO	**< €5 million**	**€ 2,000**

Micro also includes universities, regulators, public research bodies, institutions, NGOs, and non-profit organizations.

Membership information, sponsorship opportunities, partnerships, questions on fees?
Contact **Eva Markvoort** at **markvoort@5g-mag.com**

JOIN THE EFFORT

Get involved

Ways to work with us, member or not

Bring your topic or project

Don’t see your topic here? Bring it, or your project, as a member.

Early Access

Request access to Reference Tools and Testbeds still in development.

Contribute code

No membership needed: sign the CLA and open a pull request on GitHub.

Join the developers

The Dev Community on Slack and the Reference Tools mailing list.

Stay informed

Email updates, the 5G-MAG Magazine and The Handshake podcast.

Follow us

News on LinkedIn; talks and demos on YouTube.

The Media Connectivity Association

Join the Effort

Help shape what connected media looks like next and scale your projects with the industry.

GitHub

5g-mag.com

Become a member →

www.5g-mag.com/membership